

Zanzibar Low-Transmission Malaria Behavior Survey (MBS)



Zanzibar Low-Transmission Malaria Behavior Survey (MBS): Dissemination

Dr. Shija Joseph
ZAMEP

February 24, 2021



U.S. President's Malaria Initiative



Breakthrough
ACTION
FOR SOCIAL & BEHAVIOR CHANGE



Zanzibar Low-Transmission Malaria Behavior Survey (MBS): Dissemination

Mwinyi Khamis
ZAMEP

February 24, 2021



U.S. President's Malaria Initiative



Breakthrough
ACTION
FOR SOCIAL & BEHAVIOR CHANGE



Workshop objectives

1. To disseminate malaria behavior survey finding to MOH officials, development partners and other key stakeholder's

From the community/HH survey will be able share with you the followings

- i. ITNs findings
- ii. IRS findings
- iii. Community and HW findings for malaria case management
- iv. Findings for larviciding
- v. Finding for malaria surveillance

2. To collect the recommendations from the workshop participants

Zanzibar Low-Transmission Malaria Behavior Survey (MBS): Dissemination

Johns Hopkins University
Center for Communication Programs

Carol Underwood, PhD
Ruchita Pillai, MHS

February 24, 2021



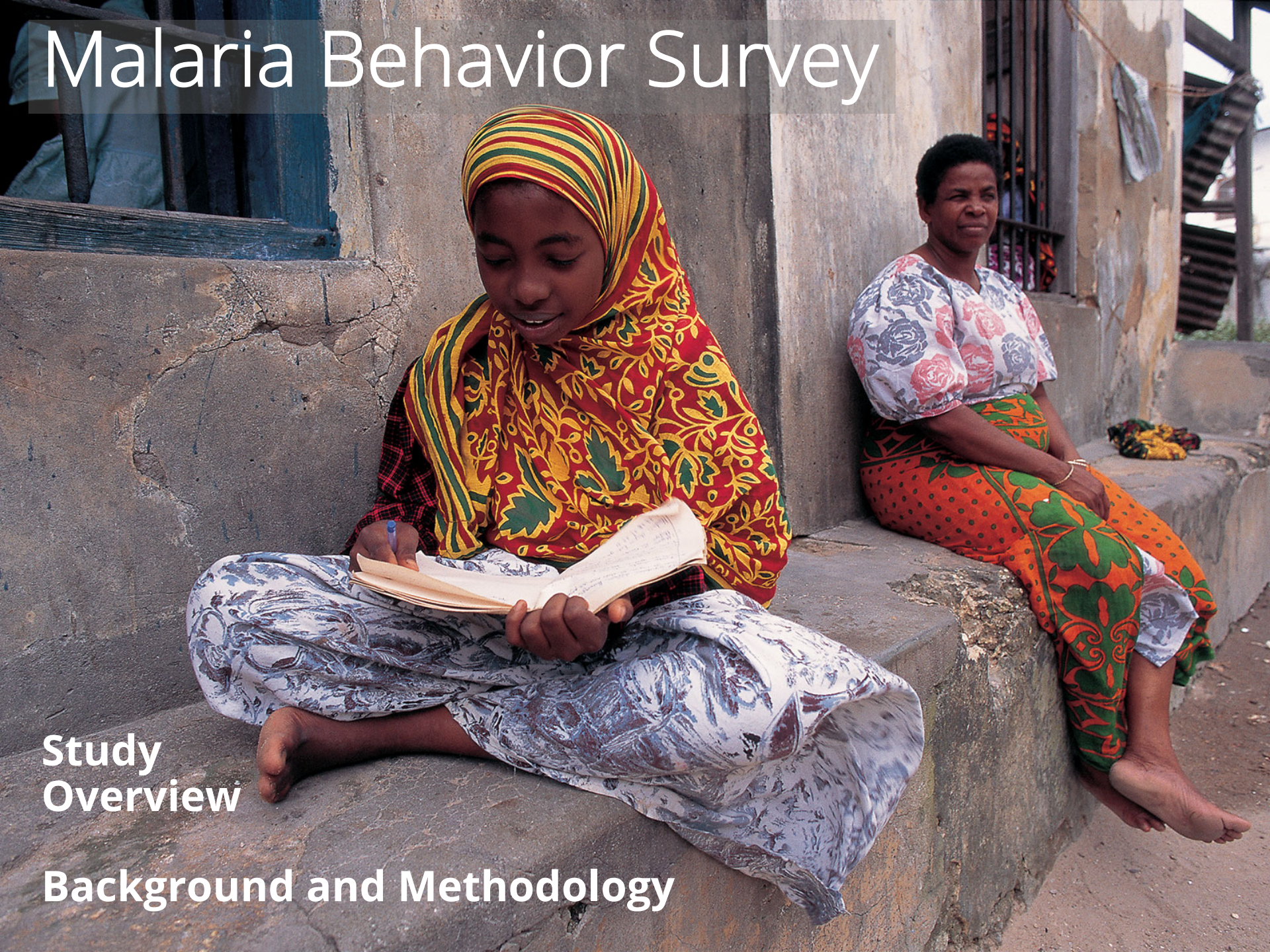
U.S. President's Malaria Initiative



**Breakthrough
ACTION**
FOR SOCIAL & BEHAVIOR CHANGE



Malaria Behavior Survey



Study
Overview

Background and Methodology

Malaria Behavior Survey

Survey Objectives

- A survey to identify and understand the demographic, ideational (e.g., knowledge, social norms, self-efficacy, intention) and contextual factors associated with:
 - bed net use and care;
 - care-seeking and treatment for malaria
 - acceptance of indoor residual spraying (IRS)
 - acceptance of larvicide
- To understand reasons for not adopting or maintaining appropriate health practices
- To assess recall of, participation in, malaria-related social and behavior change (SBC)
- To help inform and design relevant programs to improve malaria-related behaviors

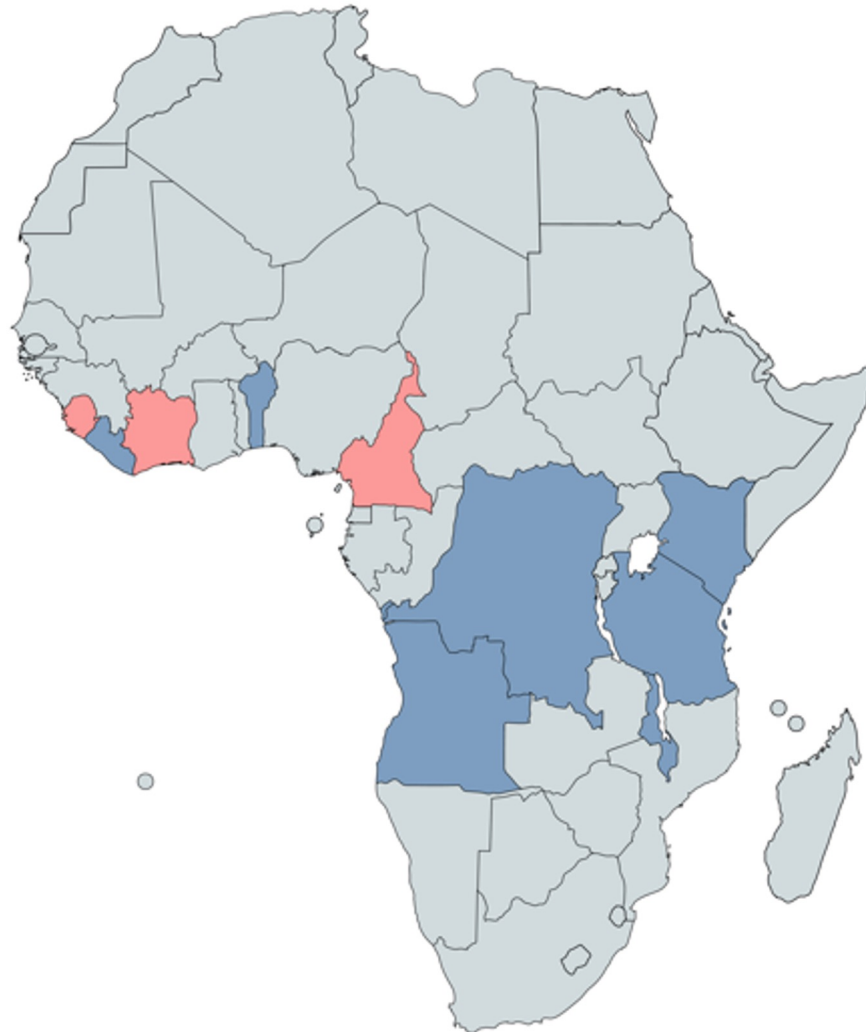
Malaria Behavior Survey Countries to date



- Côte d'Ivoire: pilot country
- Sierra Leone
- Cameroon
- Malawi
- Tanzania
- **Zanzibar** (first low-transmission survey)
- Democratic Republic of Congo
- Kenya
- Benin
- Angola

Status

- Completed
- In Progress



Conceptual Framework

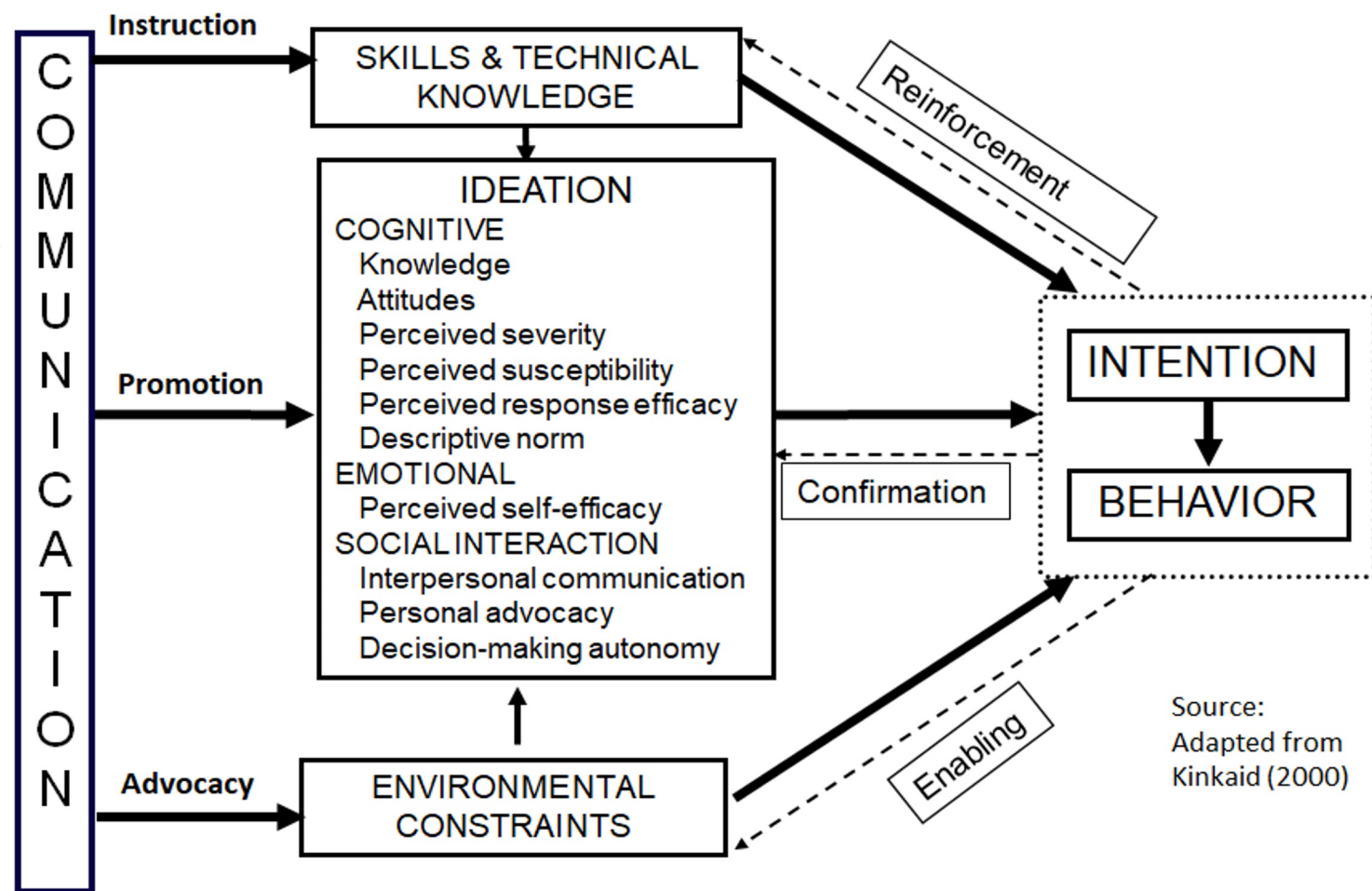
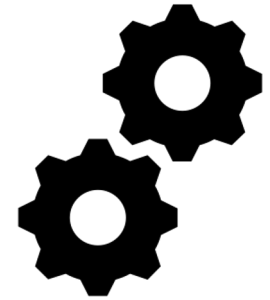


Figure 1: Ideation model of strategic communication and behavior change

Sample Description



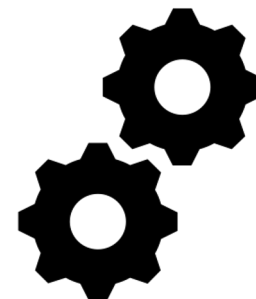
Methodology: Sampling Strategy



- Stratification by urban/rural within each zone
- Selection of enumeration areas (EAs)/clusters
 - Using probability proportionate to size
- Selection of households
 - 20-21 households per cluster
- Selection of Participants
 - In each household, interview one eligible woman and one eligible man.

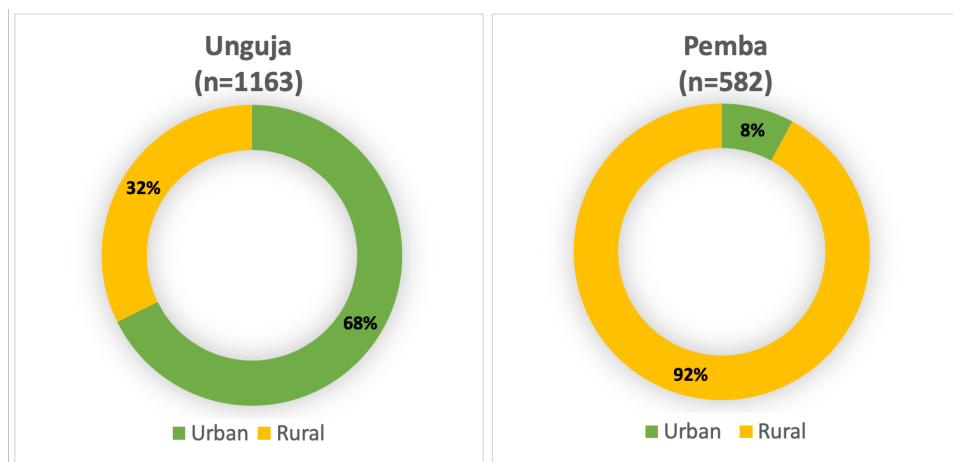


Methodology: Final Sample

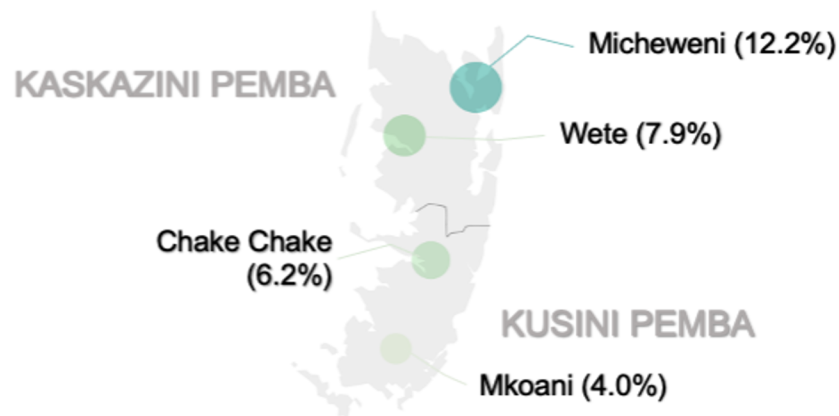


Study Sample	
	1,007 households with 5,395 members
	874 Male respondents
	871 Female respondents

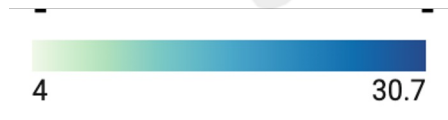
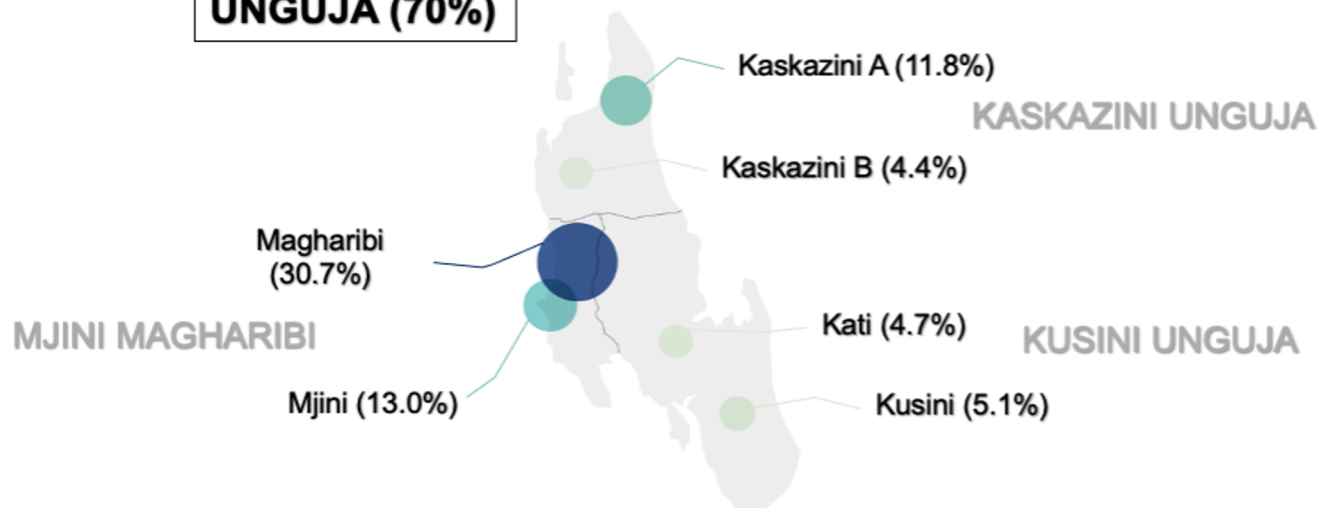
- Households
- Female respondents 15-49 years
- Male respondents 18-59 years



PEMBA (30%)



UNGUJA (70%)



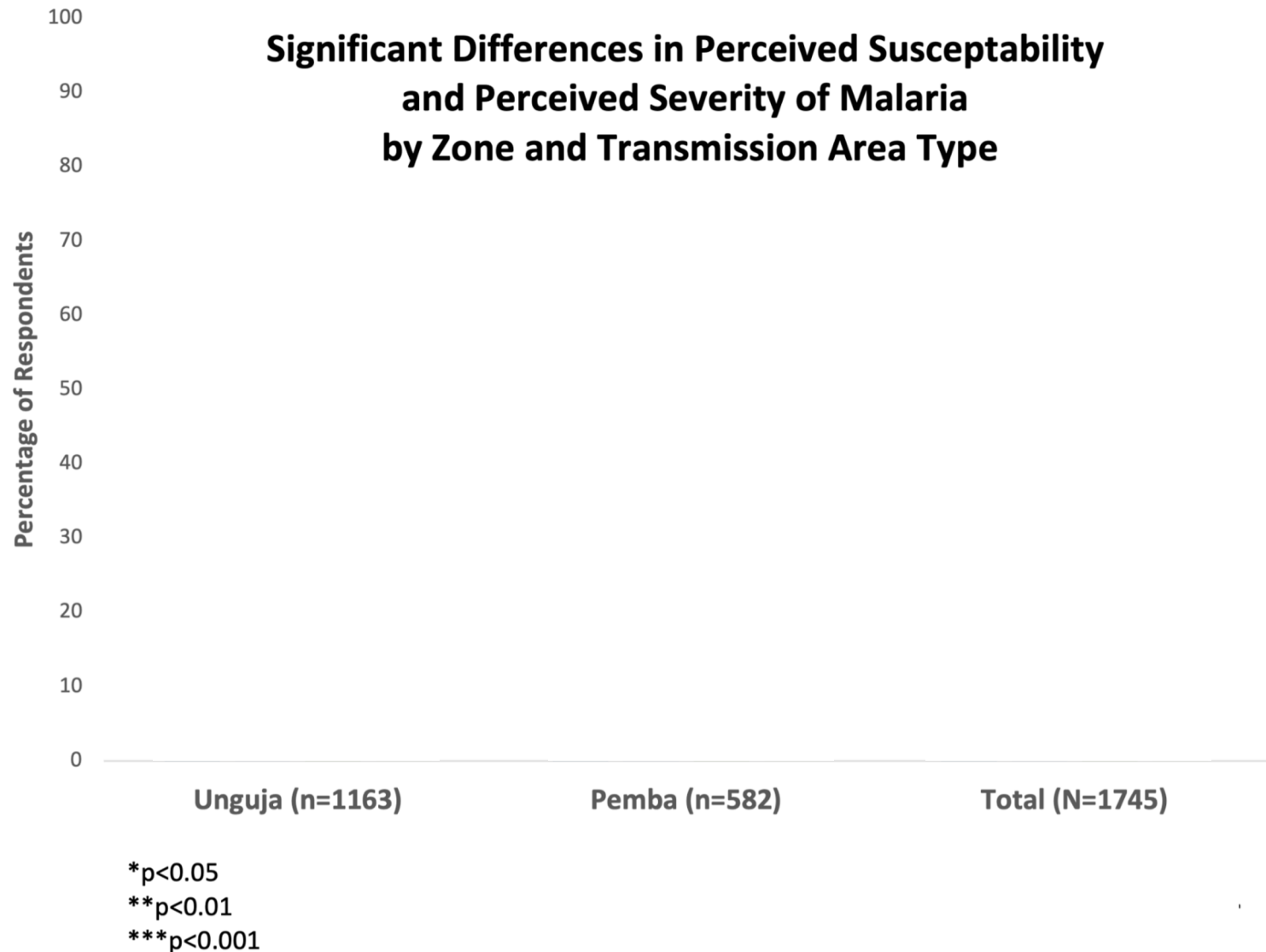
Cross-cutting Ideational Factors

- 
- A close-up photograph of three young women of African descent. The woman on the left wears a white lace headscarf and looks slightly to the right. The woman in the center wears a dark headscarf and looks forward. The woman on the right wears a white headscarf with pink floral patterns and smiles at the camera. The background is a solid teal color.
- Perceptions of susceptibility, severity, response efficacy, self-efficacy
 - Norms
 - Interpersonal communication
 - Decision-making

Ideational Factors: Malaria

Malaria-related Ideational Factors at a Glance ($\bar{x}$)	
	86% perceived equitable gender norms
	86% overall favorable perceptions of health workers
	51% perceived susceptibility
	28% perceived severity
	11% interpersonal communication with spouse/partner

Ideational Factors of Malaria





Media Consumption & Message Recall

Media Access & Consumption

(from Individual Questionnaire; N=1745)

Media Access and Consumption at a Glance		
	Radio listenership	49%
	TV viewership	42%
	Mobile phone ownership	80%

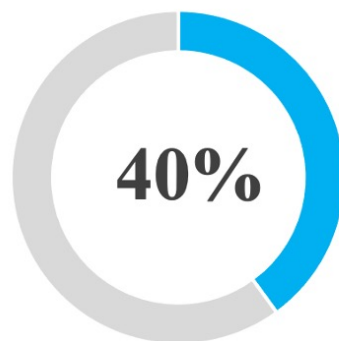
Radio/TV: Men, Unguja residents, urban dwellers, higher educational attainment, higher wealth brackets

Mobile phone: Men, older respondents, urban dwellers, higher transmission areas, at least a secondary education, upper wealth quintiles associated with significantly higher levels of ownership.

Smartphone: Ownership significantly different by zone: Unguja: 53%; Pemba 15%

Malaria Message Recall

Heard Malaria Message



Zone***:

Unguja: 44%

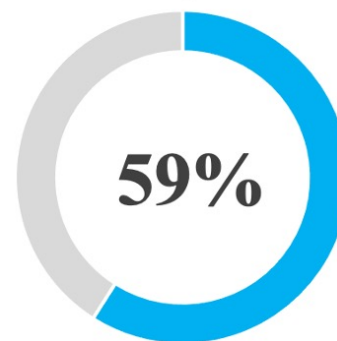
Pemba: 31%

Residence***:

Urban: 45%

Rural: 35%

Completed Campaign Slogan



Zone***:

Unguja: 66%

Pemba: 45%

Residence***:

Urban: 69%

Rural: 51%

Strategy 1: Malaria Diagnosis and Treatment – Ensure quality assured diagnosis and appropriate case management in all health facilities and at community level



Malaria Case Management

Female respondents
15-49 years;
Male respondents
18-59 years

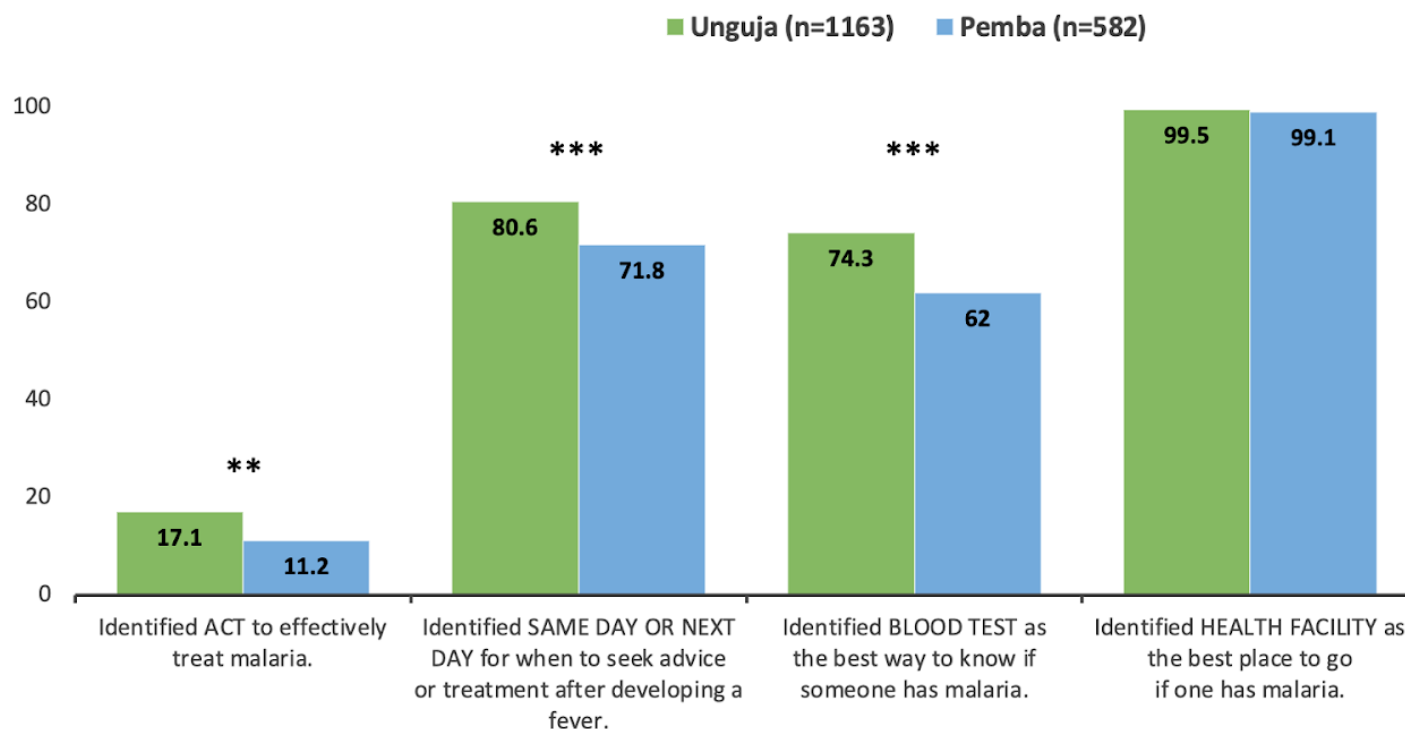
Intention to seek care for
child under five years



Carol Underwood, PhD
Johns Hopkins University

Knowledge: Care-seeking & Treatment

Figure 3.3.1 Percentage of respondents with specific knowledge of malaria care-seeking and treatment, according to background characteristics, Zanzibar 2021

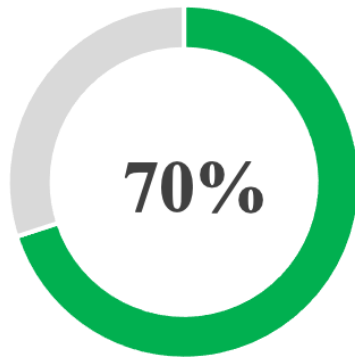


Significance levels: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

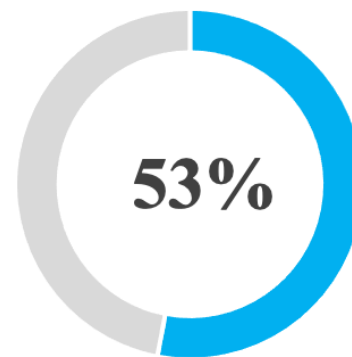
Health Facility includes the following places: Public medical sector, private medical sector and community health provider. Exclude advice or treatment from a traditional practitioner, shop, market and itinerant drug seller.

Among respondents with fever (n=164)

**Any
Care-seeking**



Tested for Malaria



Only **five** respondents reported confirmed malaria

Malaria Care-seeking and Treatment: **Facilitators**



88%

Favorable perceptions of facility-based health providers



75%

Favorable perceptions of community-based health providers

Most respondents agreed that health providers are the best people to talk about malaria (96%), that facility- (86%) and community-based (75%) providers treat their patients with respect, and that health facilities always have malaria test kits (70%) and medications (68%) in stock.

- **Gap:** Unguja residents were less sure about whether health providers (both facility- and community-based) knew how to treat malaria (54-70%) and thought providers might ask for payment for malaria tests and treatment (66-69%).
- **Gap:** For community-based health providers specifically, respondents in general were least sure about whether they knew how to treat malaria (56%) and would always have test kits (47%) and medications (46%) to treat malaria.

Malaria Care-seeking and Treatment: **Facilitators**



80%

Perceived equitable gender norms

Respondents displayed high levels of gender equitable norms regarding malaria. This suggests that most people would treat male and female children equally with respect to prevention and treatment.

- The lowest equitable perception reported (56%) was around pregnant women feeling comfortable to ask their spouse/partner to go to the health facility for a prenatal consultation.



88%

Involved in decision to seek care when has a fever

Most respondents (88%) said they either make the malaria-related healthcare-seeking decisions themselves or jointly with their partner/spouse.

Malaria Care-seeking and Treatment: **Facilitators**



93%

Perceived self-efficacy to seek testing

Respondents reported high self-efficacy to seeking malaria testing, with most respondents reporting they could afford to seek care (92%), testing and treatment (91%), would go to the health facility/provider the same day or next after developing fever (90%), and can request a blood test (92%) and make sure to finish the full dose of malaria treatment (94%).



94%

Households reported being near a health facility (public/private)

About 93% of households reported being close to a public health facility, with significant differences between Unguja (96%) and Pemba (87%). About two-thirds of the households reported being close to a private health facility. Fewer Pemba households (61%) reported being close to a private health facility than Unguja households (70%). Overall 80% of households reported being close to a pharmacy with no differences between Unguja (80%) and Pemba (79%). The difference in access to health resources between Unguja and Pemba can be explained by the fact that Pemba is 92% rural while Unguja is 68% urban.

Malaria Care-seeking and Treatment: **Limiting Factors**



69%

Recognized fever as the main symptom of malaria

About 69% of the respondents reported fever as the main symptom of malaria. If prompt care-seeking for fever is universally recommended, the gap in recognizing fever as a sign of malaria might keep people from seeking care.

- Younger respondents (<35 years), urban residents, and the two highest wealth quintiles were least likely to recognize fever as a malaria symptom.



51%

Perceived susceptibility to malaria

Only about half of the respondents perceived that they and their families were susceptible to malaria. In a low-transmission setting where malaria is increasingly rare, such observations may be prudent. Residents of higher transmission areas were significantly more likely than those living in lower transmission areas to report perceived susceptibility, which is interesting and reflects a level of knowledge about the risks they face.



28%

Perceived severity of malaria

Overall, only about a quarter of respondents — again with higher percentages in high transmission-risk areas — indicated that the consequences of acquiring malaria would be severe. Given good access to health care in Zanzibar in tandem with high levels of confidence in the healthcare system, this perception may reflect their own experiences.

Malaria Care-seeking and Treatment: **Limiting Factors**



47%

Perceived response efficacy of malaria testing

While 70% of respondents agreed a blood test to be the best way to tell if someone has malaria, less than half of the respondents thought malaria testing to be efficacious.

- **Gap:** With about 50% of respondents agreeing with the statement that a person should take malaria medication despite testing negative for it.



15%

Identified ACT as the medicine to treat malaria



37%

Perceived response efficacy of malaria treatment

Only 15% of respondents identified ACT as the medicine used to effectively treat malaria and 37% considered malaria treatment to be effective.

- **Question:** In low-transmission settings, should programs expect community members to name ACT as the medicine that effectively treats malaria? If not, messages need not focus on this ACT.
- **Gap:** Majority of respondents (82%) thought that an injection to treat malaria is more effective than malaria medicine taken by mouth.
- Unguja residents, urban residents, men, and those in higher wealth quintiles were least likely to consider malaria treatment effective.

Malaria Care-seeking and Treatment: **Limiting Factors**



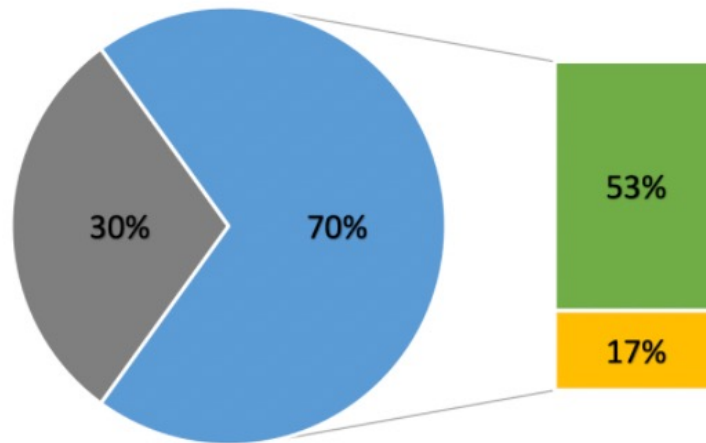
52%

Perception that most people seek prompt care

Only about half the respondents thought that people in their community sought prompt care at a health facility for a fever.

- Residents of Unguja, urban areas and high transmission areas, and those in the two highest wealth quintiles were less likely to think their community members seek prompt care.

Care-seeking Behavior for Fever



% of those with fever who sought care

% who sought care the same day or next after the onset of fever

% who sought care sometime after the 48 hours from onset of fever

% who did not seek care for their fever

Among respondents who had a fever in the two-week period prior to the survey, 70% sought care for the fever, yet only 75% of those individuals did so within the recommended time frame (the same day or next day after onset of fever).

- Care-seeking was significantly lower among rural respondents, which could reflect more difficulty accessing a clinic.
- Close to three-fifths of respondents reported that most people in their communities seek prompt care upon developing a fever (descriptive community norm). These percentages could be improved with appropriate interventions across social ecological levels.
- Among those who sought care, about half were tested for malaria at the clinic; five individuals tested positive.

Mass Drug Administration (MDA)



Malaria Care-seeking and Treatment - Mass Drug Administration: Limiting Factors



16%

Aware of Malaria MDA programs

Only 16% of respondents reported being aware of MDA programs for malaria. Pemba residents, men, older respondents (35+ years), and those with primary or higher education were more likely to be aware of MDA programs than their group counterparts.

Malaria Care-seeking and Treatment - Mass Drug Administration: Facilitators



70%

Perceived supportive community norms regarding MDA

Pemba residents and those in the lower wealth quintiles were more likely to perceive supportive community norms around MDA programs. About 66-67% of respondents thought that most community members and most community leaders would accept antimalarials if offered through a mass community distribution.



69%

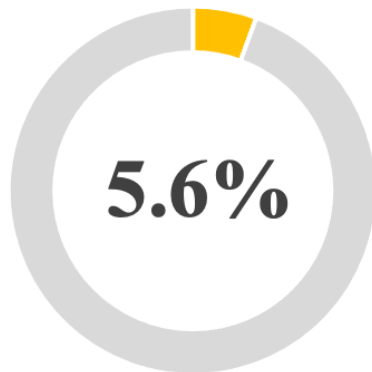
Willingness to accept antimalarials from a MDA program despite not feeling sick

Among all respondents,⁴ almost 70% stated that they would accept antimalarials for malaria treatment as part of MDA despite not feeling unwell. The willingness to participate was higher among urban respondents and those living in higher transmission areas when compared to their group counterparts.

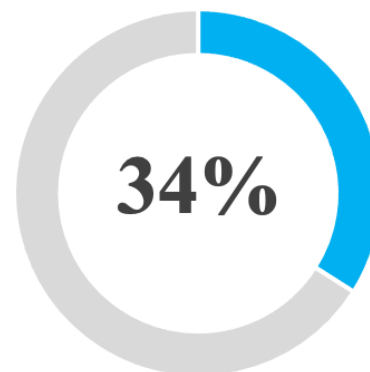
MDA Coverage

Reported receiving antimalarials through an MDA program:

Among All Respondents



Among respondents aware of MDA



Statistically significant difference by zone: Unguja: 42%; Pemba: 21%

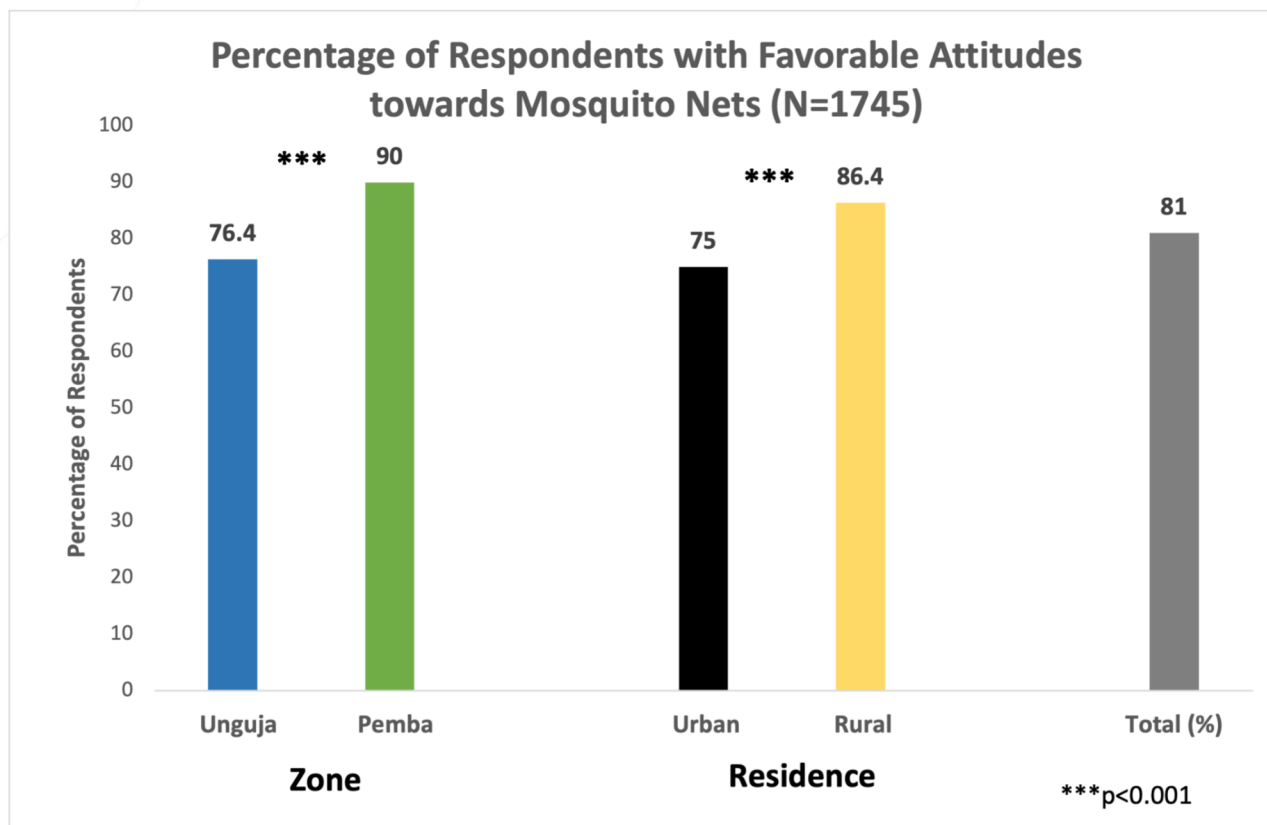


Strategy 2: Integrated Malaria Vector Control -
Increase appropriate vector control measures to
the population at risk of malaria

Mosquito Net/ Insecticide-Treated Net Use



Ideational Determinants of Net Use



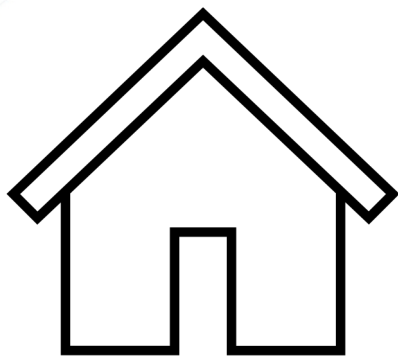
It is not easy to bring a net when I spend the night away from home
(65% agreed with the statement)

Treated mosquito nets attract bed bugs and other insects
(59% agreed with the statement)

Ideational Determinants of Net Use

- ❖ Response efficacy: 67% of respondents agreed with the statement: *My chances of getting malaria are the same whether or not I sleep under a mosquito net*
- ❖ But most (86%) agreed that sleeping under a mosquito net every night is the best way to avoid getting malaria.
- ❖ Most respondents (86%) also had high perceived self-efficacy for net use, with **sleeping outside** (55%) and **sleeping/working away from home** (60-71%) being two instances of lower perceived self-efficacy to net use.
- ❖ Community norm: *at least half of the community members sleep under net every night*
 - more respondents from **Pemba** (87%) reported supportive net use community norm than from **Unguja** (62%)
 - more **Rural** respondents (76%) reported supportive net use community norm than **Urban** respondents (64%)

Household Net Access



Number of HHs in the survey

N=1007

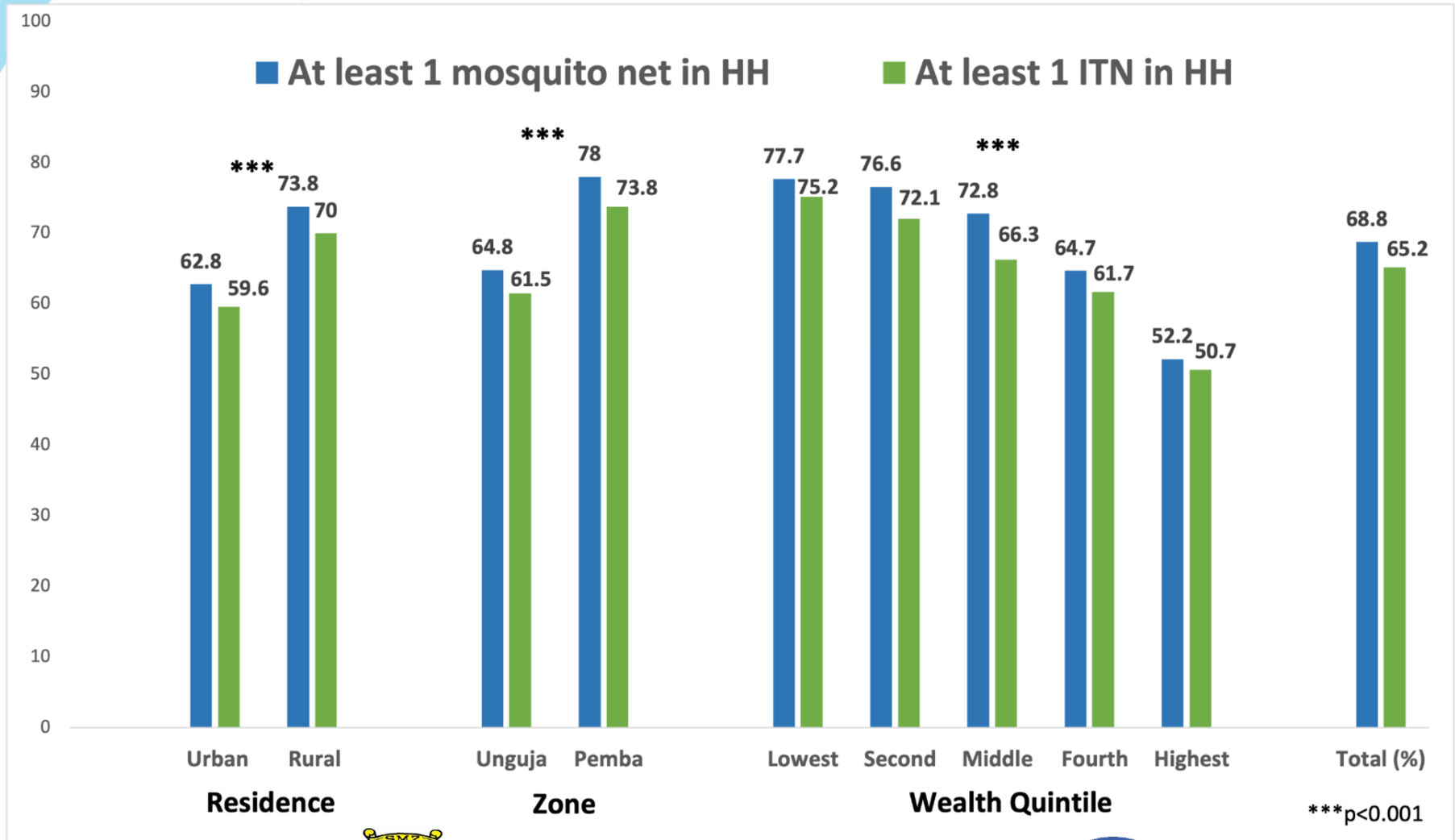


65%

Number of nets in each household (Net Access)

- At least one mosquito net (treated/untreated)
 - At least one ITN

Household Net and ITN Access



Use of Available Nets in HH



How many nets in the house (Net Access)

n=1289 nets

in N=1007 HHs

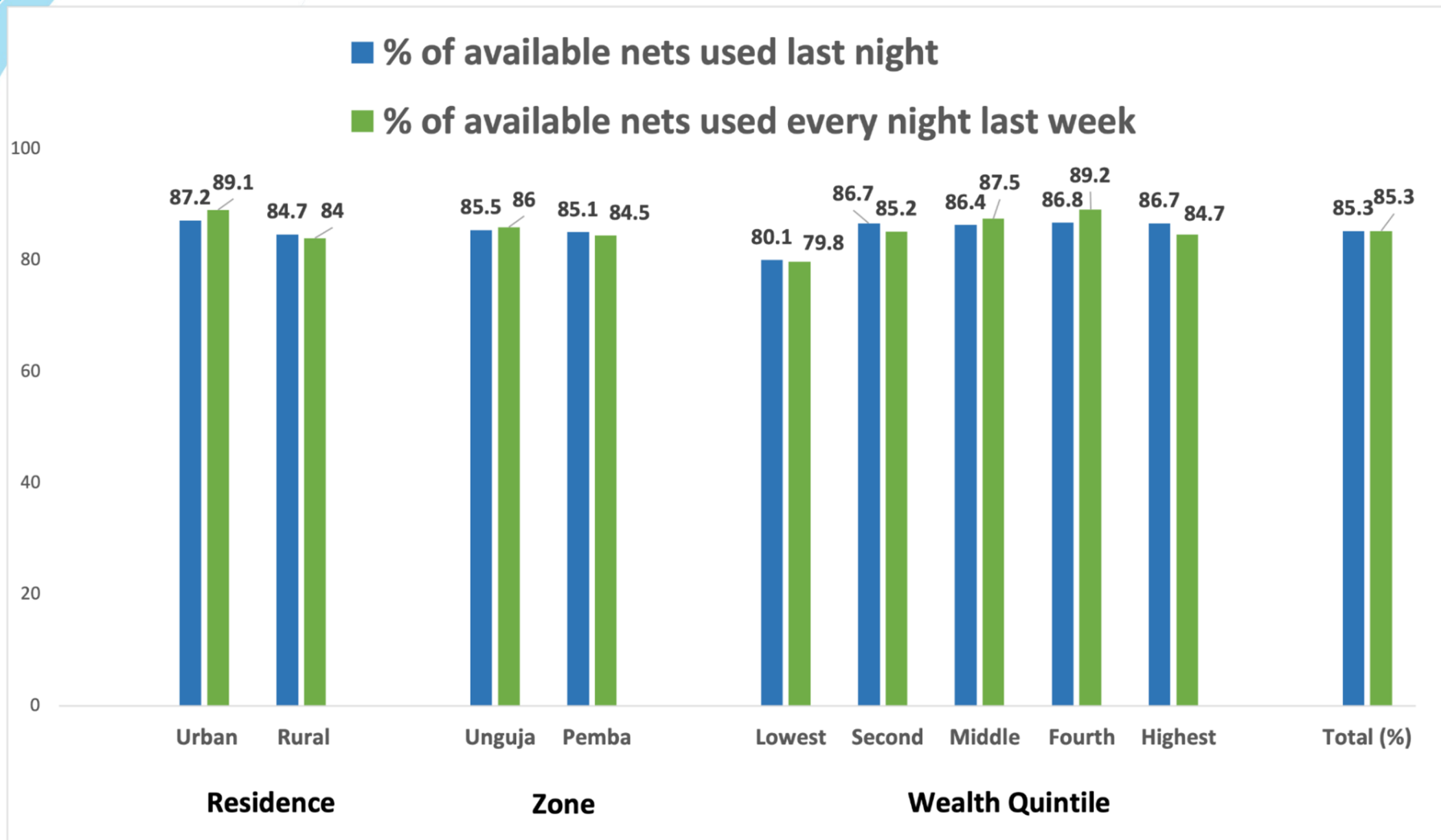


85%

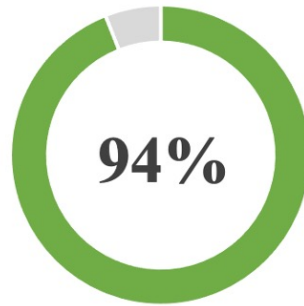
How many of the available nets were used

- The night before
- Every night the week before

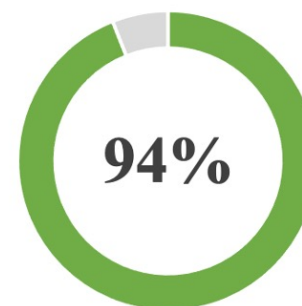
Use of Available Nets (1289 nets from 1007 HHs)



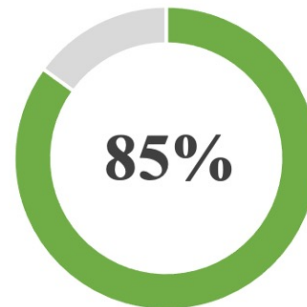
% of nets that were
ITNs



% of nets obtained
free of charge

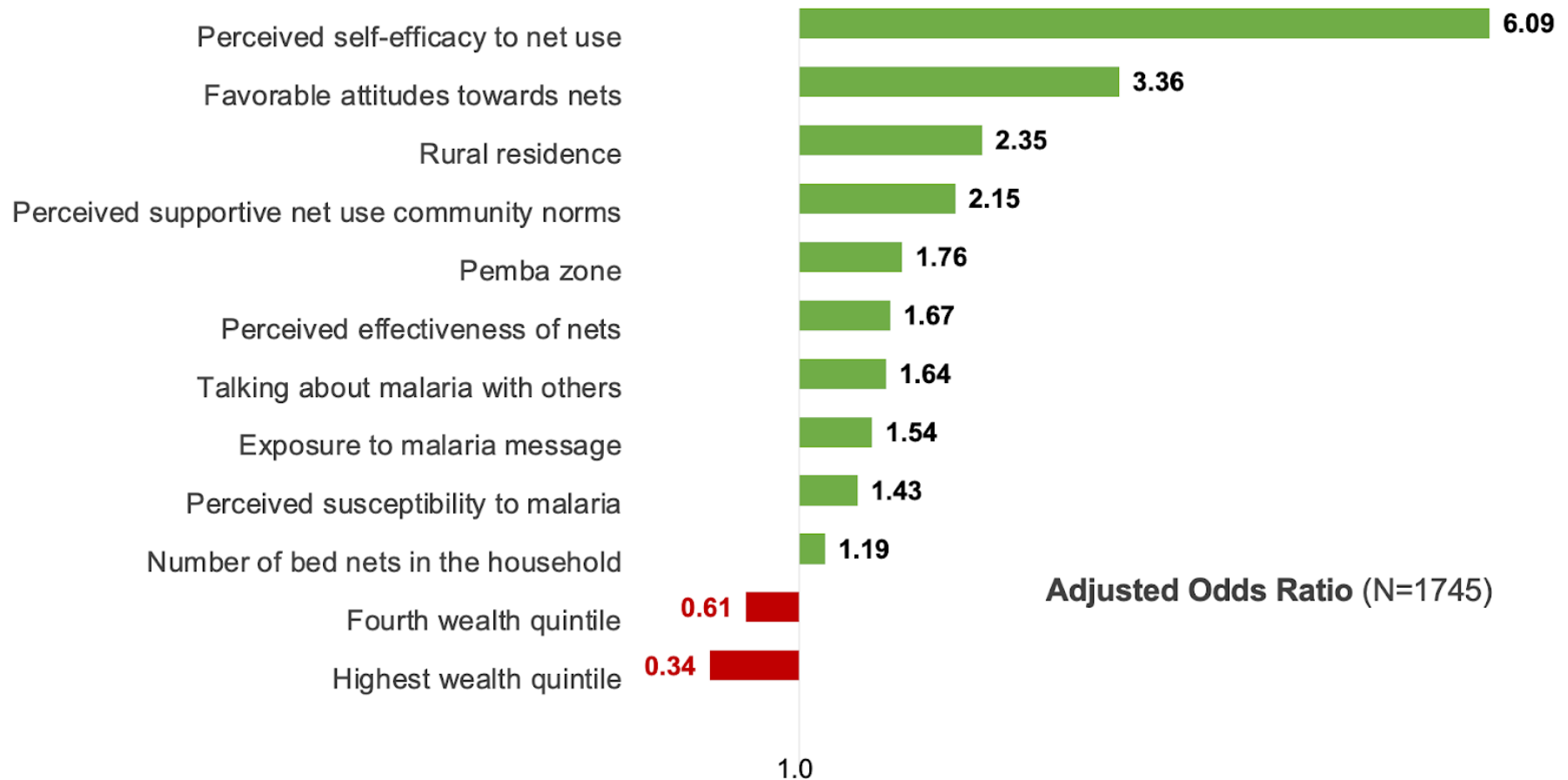


% of bed nets used
consistently



Yet only **37%** of respondents reported consistent use
(net used every night of the week before the survey).

Significant Determinants of Consistent Net Use



Malaria Vector Control - Mosquito Nets: Facilitators



91%

Malaria prevention knowledge by bed net use (treated/untreated)

Majority of respondents identified mosquito nets or insecticide-treated nets as a major malaria prevention method. Only slight but still significant differences existed in respondents' knowledge between urban (93%) and rural (90%) residents, and by levels of educational attainment (89-92% linear trend with increase in education).



81%

Favorable attitudes towards bed nets

While most respondents reported favorable attitudes towards bed nets, respondents in the highest wealth quintile (71%), urban areas (75%), and of ages 15-24 years (76%) were least likely to have these favorable attitudes.

Statements where respondents reported the most unfavorable attitudes:

- Only about half the respondents or less thought it was easy for them to: unfold a net and cover a sleeping area every night (50%), use a net while traveling or working away from home (49%), bring a net while sleeping away from home (35%), and use a net while sleeping outside (43%)
- Half the respondents reported disliking sleeping under a net when the weather is warm
- About 52% reported feeling uncomfortable sleeping under a net on account of the smell of insecticide
- More than 70% thought more expensive bed nets were more effective than cheaper or free bed nets
- More than half the respondents thought sleeping under a net caused low sex drive in men (57%) and that treated nets attracted bed bugs and other insects (59%)

Malaria Vector Control - Mosquito Nets: Facilitators



86%

Perceived self-efficacy to use bed nets

Overall self-efficacy to use bed nets was high among respondents. The lowest self-efficacy was reported in the following instances:

- Sleeping under a net most nights when they are *sleeping* away from home (71%)
- Sleeping under a net most nights when they are *working* away from home (60%)
- Using a mosquito net most of the time when they are *sleeping outdoors* (55%)



76%

Perceived equitable gender norms toward net allocation

Those with a primary level of education (70%) and those between the ages of 15-24 years (69%) were the only two groups least likely to report equitable gender norms for net allocation.



70%

Perceived consistent bed net use by community members

The perception that most community members consistently (every night) used a bed net was common among respondents. But respondents from Unguja (62%), from urban areas (64%), with secondary or higher education (65%), and in the middle to highest wealth quintiles (62-69%) were significantly less likely to have this perception compared to their counterparts.

Malaria Vector Control - Mosquito Nets: Limiting Factors



65%

Households with access to at least one insecticide treated net

70% of households reported having access to at least one bed net and 94% of those bed nets were reported to be ITNs.

- Access to at least one ITN was significantly lower for households in urban areas, Unguja, and the two highest wealth quintiles compared to their counterparts.



37%

Respondents reporting using an ITN every night

About 85% of nets were reported to have been used every night the week before the survey but fewer than four in ten respondents reported that they consistently used their nets (every night of the week preceding the survey). In contrast to findings from the standard MBS across multiple countries, where relative privilege is typically associated with mosquito net use, in Zanzibar consistent mosquito net use was significantly higher in rural areas, Pemba, and the two lowest wealth quintiles.

- Ideational factors — particularly perceived self-efficacy to use nets, favorable attitudes toward net use, and supportive descriptive community norms — were important enablers of consistent use.

Malaria Vector Control - Mosquito Nets: Limiting Factors



43%

Perceived response efficacy of bed nets

Majority of respondents agreed that sleeping under a mosquito net every night is the best way to avoid getting malaria (86%) and that using a mosquito net while sleeping outside would reduce the chances of getting malaria (89%). But only one in three or less disagreed with the statements: “mosquito nets prevent mosquito bites only when used on a bed” (30%) and “my chances of getting malaria are the same whether or not I sleep under a mosquito net” (33%).

- Respondents from urban areas, higher transmission areas, and the highest two wealth quintiles were least likely to think mosquito nets as effective in preventing malaria.



43%

Perceived coworker approval of bed net use when working away from home

Of those who reported working away from home, 62% thought that their coworkers used a bed net every night when working away from home but only 43% thought that their coworkers would approve of consistent bed net use when working away from home.

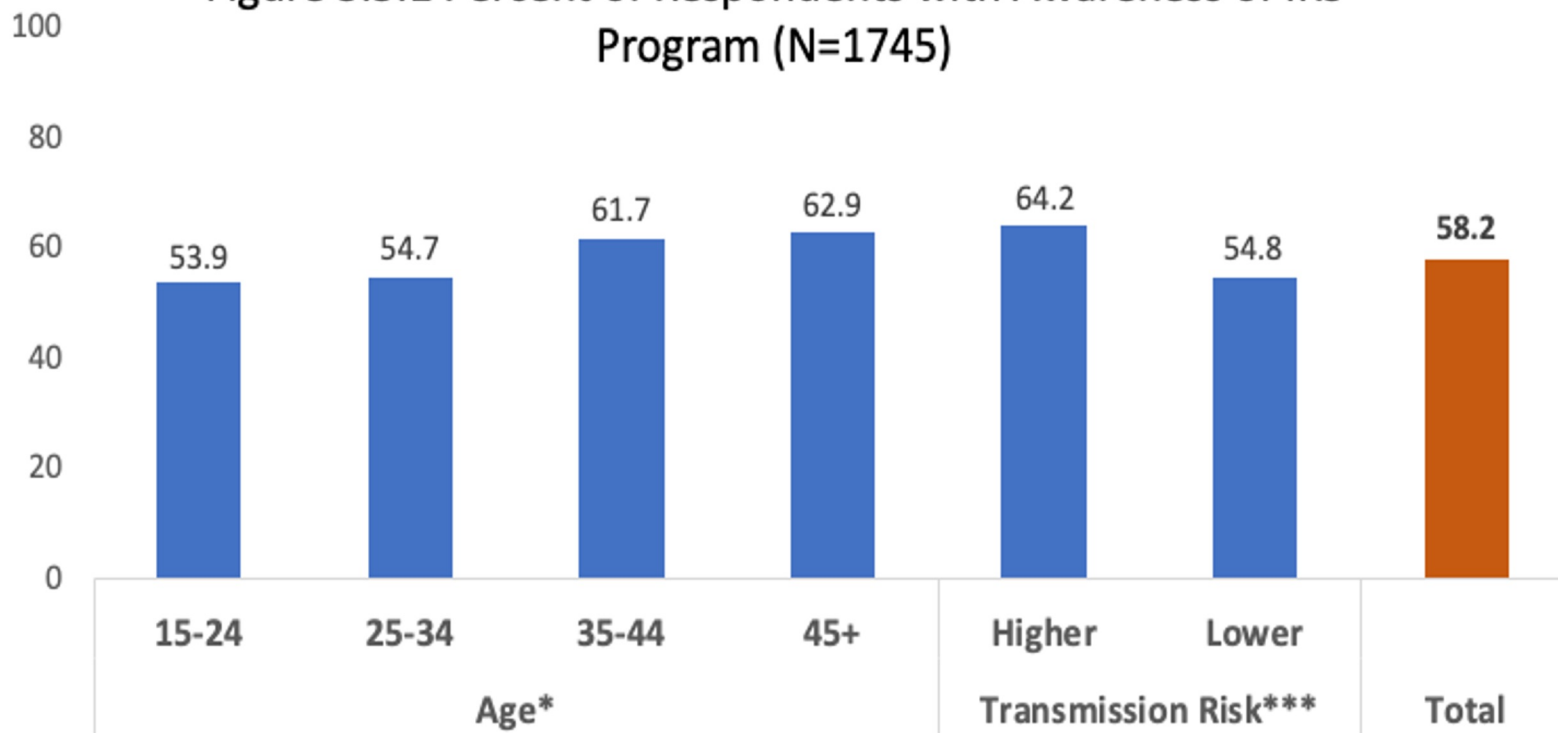
Indoor Residual Spraying (IRS)



Ruchita Pillai, MHS
Johns Hopkins University

IRS Awareness

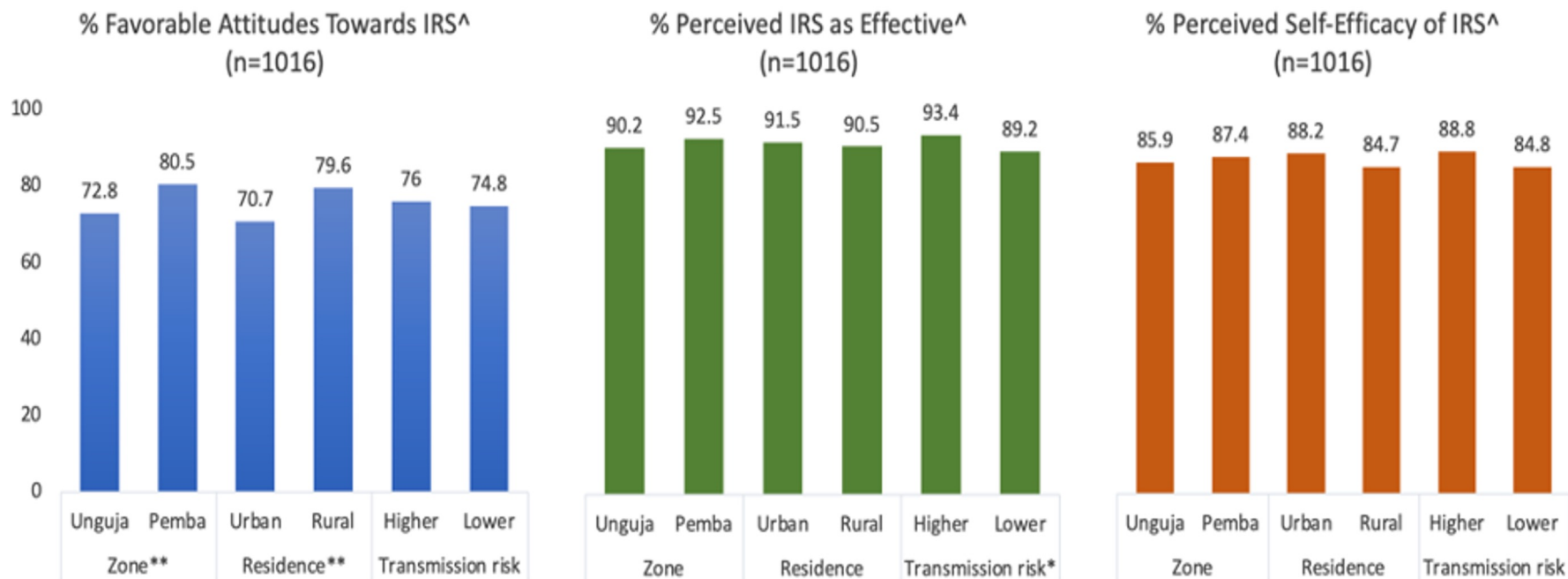
Figure 3.5.1 Percent of Respondents with Awareness of IRS Program (N=1745)



Significance level: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

Note: Background characteristics including zone, sex, residence, education level, and wealth quintile were not significant

Figure 3.5.2 IRS Attitudes, Response-Efficacy, and Perceived Self-Efficacy



Significance level: *p<0.05; **p<0.01; ***p<0.001

Notes: ^Only those respondents who were aware of the IRS program were asked these set of questions (n=1016)

Ideational Factors of IRS

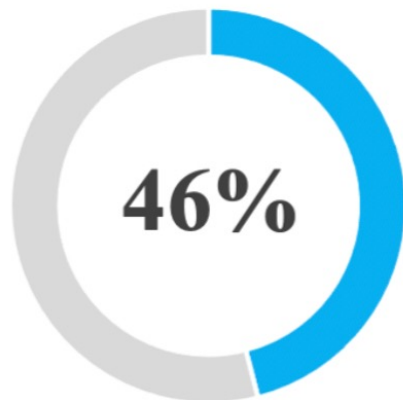
Attitudes (limiting)

- *Many people develop skin problems (rashes, itching) after the walls inside their houses are sprayed with insecticide*
 - **60% agreed** with this statement
- *People have problems with bugs/bed bugs after the walls are sprayed*
 - **48% agreed** with this statement
- *It bothers me to leave my possessions outside of my house while my walls are being sprayed*
 - **44% agreed** with this statement

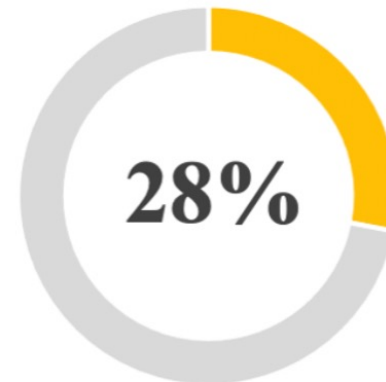
Response Efficacy (supporting)

- *Spraying the inside walls of a house is an effective way to prevent malaria*
 - **94% agreed** with this statement
- *People who live in houses that have been sprayed are less likely to get malaria*
 - **92% agreed** with this statement

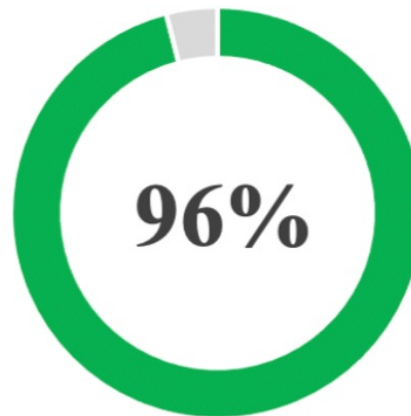
Pemba households reported IRS was offered



Unjuga households reported IRS was offered

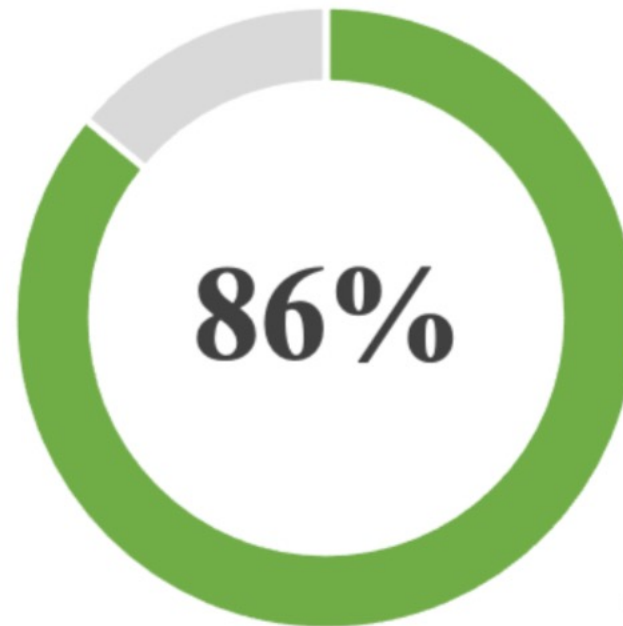


Accepted IRS among those offered



Among those
aware of IRS
(n=1016)

Would accept IRS if offered



N=1745

All respondents; N=1745. Only significant difference by transmission risk: higher transmission risk areas: 89%; lower transmission risk areas: 84%.

Malaria Vector Control - Indoor Residual Spraying: Facilitators



75%

Favorable attitudes towards IRS

Of those who reported being aware of IRS, 75% reported favorable attitudes towards it. Rural respondents (80%) were more likely than their urban counterparts (71%) to have these favorable attitudes.

- **Gap:** Only 41% of respondents disagreed with the statement that many people develop skin problems after having their walls sprayed
- About 48% still believe that people have problems with bed bugs and other insects after their walls are sprayed
- Half the respondents also expressed discomfort in leaving their possessions outside their house while the walls are being sprayed



91%

Perceived IRS as effective

Nearly all respondents considered IRS as an effective method to prevent malaria. Only a slight but statistically significant difference existed between higher transmission area respondents (93%) and lower transmission area respondents (89%).

- **Gap:** A common misconception was noted where about 63% of respondents thought there is no need to sleep under a net once their house is sprayed.

Malaria Vector Control - Indoor Residual Spraying: Facilitators



86%

Perceived self-efficacy regarding IRS

While majority respondents reported self-efficacy regarding IRS, there was a significant difference among respondents from Unguja (90%) and Pemba (99%) reporting that they could sleep in the house on the night it is sprayed. In Pemba, more residents in the higher transmission areas (94%) reported self-efficacy towards IRS than those in lower transmission areas (82%).



85%

Willingness to accept IRS

Among all respondents,⁵ 85% stated that they would accept IRS if offered. This reflects broad trust among Zanzibaris in governmental structures. Only respondents from higher transmission areas (89%) were significantly different and more likely to accept IRS than their lower transmission counterparts (84%).

Malaria Vector Control - Indoor Residual Spraying: Limiting Factors



58%

Aware of IRS programs

58% of respondents reported awareness of an IRS program in their community. Younger respondents (15-34 years) and residents of lower transmission areas were less aware of IRS programs than their counterparts.



33%

Households reporting an IRS-related visit to their dwelling

Only 33% of households reported a visit from anyone asking to spray their dwelling in the 12 months before the survey. More households in Pemba (46%) reported such a visit than households in Unguja (28%).

Of those households who were approached for IRS, 96% accepted it for their household in the 12 month period. No significant differences in household IRS acceptance were noted by residence area, transmission area, or wealth quintile, indicating reach and coverage of IRS programs being the main factor limiting the proportion of households that are sprayed in Zanzibar.

A large crowd of African women, many wearing colorful patterned headwraps, are gathered together. The women are looking in various directions, some towards the camera and others away. The headwraps feature various patterns and colors, including purple, orange, red, and blue. The women are dressed in traditional African attire. The background is filled with more women, creating a sense of a large gathering.

Larval Source Management: Larviciding

Malaria Vector Control - Larviciding: Facilitators



92%

Favorable attitudes towards larviciding

Majority respondents reported favorable attitudes towards larviciding, with some slight, significant differences between urban (94%) and rural (90%) residents, and between higher (95%) and lower (90%) transmission areas.



60%

Perceived supportive community norms regarding larviciding

While 58% of respondents thought their neighbors would approve of their community treating the water near their homes with mosquito larvicide, only 31% thought that most villages around their community actually treat the water near their homes.

- Unguja residents and urban dwellers were more likely to think that villages around their community treat the water with larvicide
- While Pemba residents were more likely to think their neighbors would approve of having water treated with larvicide

Malaria Vector Control - Larviciding: Limiting Factors



29%

Aware of larviciding programs

29% of respondents reported being aware of larviciding programs.

- More Unguja residents (40%) than Pemba residents (7%) were aware of larviciding
- Urban residents (42%) were more likely to be aware of larviciding than rural respondents (17%)
- Those with primary level of education (18%) and in the lowest wealth quintile (14%) were significantly less likely to be aware of larviciding than their group counterparts.



27%

Perceiving larviciding as effective

About 82% of respondents agreed that treating water bodies with larvicide once a week throughout the year is a good method to prevent malaria in their communities. But only 27% disagreed with the statement that their chances of getting malaria are the same regardless of water bodies in the area being treated with larvicide.

- Urban respondents, Unguja residents, and those with no education attainment were less likely to see larviciding as effective.



6%

Households reporting a larviciding program visiting their community

Only 6% of households reported a larviciding program visiting their community in the 12 months before the survey. Of the households who reported such a visit, 86% reported accepting larviciding.

- Almost all of the Unguja respondents (95%) accepted larviciding while only 63% of Pemba respondents did the same (note: small sample total of 56 households)
- High rates of reported acceptance highlights that the gap remains at the reach and coverage of larviciding programs in Zanzibar.

An elderly man with grey hair, wearing a white traditional cap and a white long-sleeved shirt, is shown in profile. He is resting his chin on his right hand, looking thoughtfully at a blue wooden door. The door has a circular metal handle and shows signs of wear and peeling paint. The background is a bright blue wall.

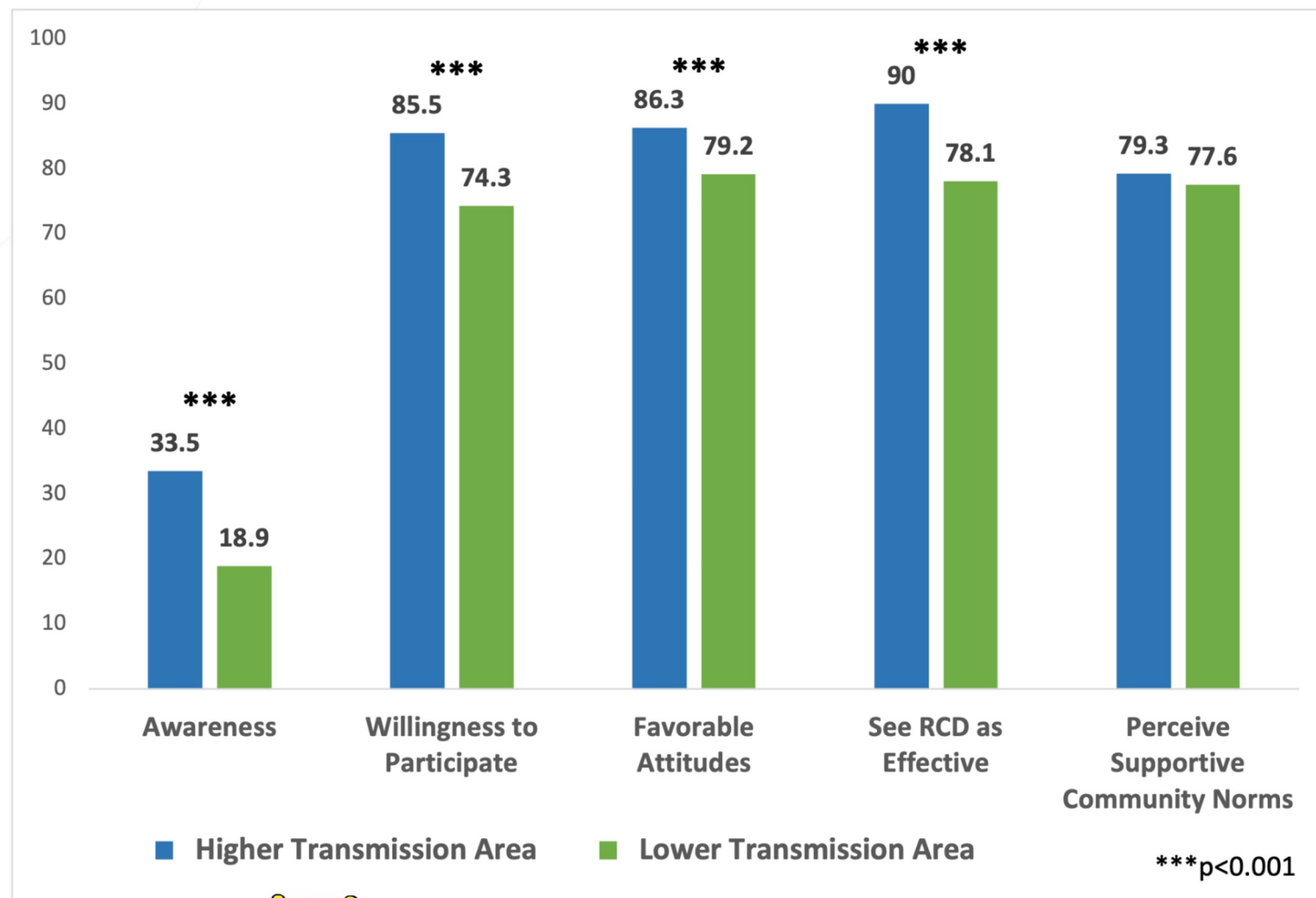
Strategy 3: Surveillance, Monitoring and Evaluation:
Actively investigate and classify 100 percent of all
confirmed cases of malaria and initiate entomological
surveillance in 100 percent of malaria foci



Reactive Case Detection (RCD)

Households and individuals within a specific area are tested and treated for malaria at their home/community when malaria cases in that area become detected, with the goal of preventing further malaria transmission by identifying additional infections, whether symptomatic or asymptomatic, and treating them promptly.

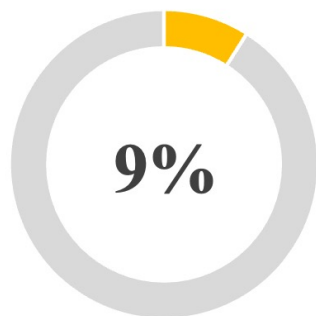
Ideational Factors of RCD



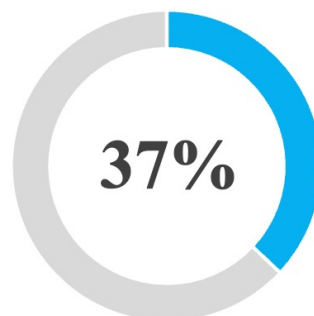
Reactive Case Detection Coverage

Reported health worker/provider visited them for RCD:

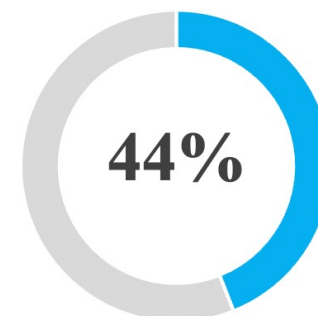
Among All Respondents



Among respondents aware of RCD



Were tested for malaria during RCD



Seven people tested positive out of the 69 who were tested during an RCD visit (10%)

Zone*:**

Unguja: 31% Pemba:
46%

Transmission Risk*:**

Higher: 48% Lower: 24%

Surveillance, Monitoring & Evaluation - **Reactive Case Detection: Facilitators**



82%

Favorable attitudes towards RCD

Among all respondents, almost eight in ten reported favorable attitudes towards RCD. Higher transmission area respondents (86%) and male respondents (84%) were more likely than their low transmission (79%) and female counterparts (79%) to have these favorable attitudes.

- **Gap:** Less than half of the respondents thought that those who feel healthy should still get tested for malaria (44%) or accept treatment (47%) following a positive test when those services are provided by a health provider visiting their home/community.
- **Gap:** Only one in three respondents stated they would trust the people who conduct unsolicited visits to test for malaria.



84%

Perceived RCD as effective

Majority of respondents thought testing households and communities after a malaria case has been identified is an effective way to identify new cases and such programs would have the appropriate treatment available when new malaria cases are identified. Overall more Pemba residents (90%) thought RCD as effective than those from Unguja (82%).

- Those from higher transmission areas (90%) were more likely to believe in RCD's efficacy than lower transmission area respondents (81%).

Surveillance, Monitoring & Evaluation - **Reactive Case Detection: Facilitators**



83%

Perceived self-efficacy towards RCD

Most respondents reported that they could make sure that their spouse/partner accepts the invitation for testing when asked by a health worker/provider who comes to their home and accept the invitation for testing when asked by a health worker/provider who comes to your home without getting anyone else's permission.

- Urban respondents (87%) and those from higher transmission areas (90%) were more likely to report self-efficacy towards these actions than rural (78%) and lower transmission area (78%) residents.



78%

Willingness to participate in community/household RCD (testing and treatment) despite not feeling sick

Among all respondents,⁶ almost 80% stated that they would get tested (80%) and treated (83%) for malaria as part of RCD despite not feeling unwell. The willingness to participate was higher among urban respondents and those living in higher transmission areas when compared to their group counterparts.

Surveillance, Monitoring & Evaluation - Reactive Case Detection: Limiting Factors



24%

Aware of RCD programs

Less than a fourth of respondents reported being aware of RCD programs.

- Those in higher transmission areas (34%) were more likely to be aware of RCD than those in lower transmission areas (19%).



45%

Reporting being tested as part of an RCD visit to their home

Of the 37% of respondents who reported a health worker/provider visiting their house to test them or someone in their household for malaria, 45% reported getting tested for malaria during that visit.

- More Pemba respondents reported being visited for RCD (46%) and getting tested during the visit (55%) than their Unguja counterparts (visited 31%; tested 35%).
- Those living in higher transmission areas (48%) were more likely to report having an RCD visit to their home than those from lower transmission areas (25%).
- In Unguja, there was a slight but significant trend around those in lower wealth quintiles being less likely to report an RCD visit than those in the higher wealth quintiles.



Research Implications

Research Implications

- More research:
 - Health facility assessment – recommended to further understand why only 53% were tested when they had a fever, when 70% sought care for that fever
- Unguja and Pemba are very different – more understanding is needed on several factors to make sure programs and policy make equitable differences across the two zones

Recommendations

From the MBS findings and
from the Data Interpretation Workshop



Strategy 1: Malaria Diagnosis and Treatment – Ensure quality assured diagnosis and appropriate case management in all health facilities and at community level

- Monitoring and enforcement of malaria drug prescription so only cases confirmed with a positive malaria test are treated with ACTs – especially in the private sector
- Low trust in a negative test of malaria
- Low perceived effectiveness for malaria treatment
 - Preference for injection over pills
- Supportive supervision/monitoring that malaria tests are administered when patient presents with fever and other symptoms of suspected malaria
 - Also, more understanding of how providers make the decision to test for when a malaria tested is needed is important – identify gaps where people might be missed/not tested
- Proper counseling from providers
 - Dose completion: discontinuing meds when feeling better
 - What a negative test means: requesting and taking ACTs even with a negative test
- If mass drug administration is considered, it would be well-received by Zanzibaris – there is high acceptance/willingness to participate already

Strategy 2: Integrated Malaria Vector Control - Increase appropriate vector control measures to the population at risk of malaria

- ITN net access improved – ~35% of HHs reported having no nets (ITNs or untreated) – Mass Distribution (July/August 2021)
 - 85% of nets are reported to be used – highlighting an access issue when only 37% of respondents reported consistently sleeping under a net
 - Will be important to look at the MIS data for this year to see how the Mass Distribution from last year changed these indicators of access and use
- For community-based programs, such as IRS or Larviciding – announcing program visit and purpose to the community/people ahead of time = building trust and increasing knowledge/awareness
- High acceptance and willing to participate in IRS and Larviciding programs
 - Great foundation for achieving higher coverage of these programs, should they be scaled up
- More information might be needed to be provided at the community-level to make people aware of the newer/infrequent programs such as IRS and larviciding

Strategy 3: Surveillance, Monitoring and Evaluation: Actively investigate and classify 100 percent of all confirmed cases of malaria and initiate entomological surveillance in 100 percent of malaria foci

- Gaps with Reactive Case Detection
 - Not everyone reported being tested despite household being visited: increased monitoring and understanding why people are being missed is needed
- Determining if workplaces are actual risk sites and looking at RCD for workplace rather than at the household level where applicable
- Not only to track malaria cases but also efficient monitoring for addressing gaps within the implementation of programs

SBCC Recommendations

- Low susceptibility, low severity towards malaria
- Low IPC about malaria – spouse/partner or friends/family past six months
- Audiences – younger generations 15-24 least likely to know/be aware/feel risk of malaria than older generations – possible older generations know about malaria from previous
- Injection more effective/not finishing drugs when they feel better – not know ACT and its effectiveness or trusting it
- SMS focused on ideational factors can be beneficial, while keeping in mind the clear need to reach those who will be left out by other means – communicating to people when programs are coming and what they are doing – add on effectiveness monitoring/tracking
- Community and facility-based interventions – getting to the people rather than just media -> get to people

Asante Sana! Shukran!

Dr. Shija Joseph
Mwinyi Khamis
ZAMEP

Carol Underwood, PhD
Ruchita Pillai, MHS
Breakthrough ACTION
Johns Hopkins University

Email: info@breakthroughaction.org

